

Game Theory for Data Science

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Introduction to Game Theory

Game Theory is a mathematical framework for analyzing strategic interactions among rational decision-makers where each participant's outcome depends on the actions chosen by all participants.

It provides systematic techniques for analyzing strategic interactions among rational decision-makers.

Applications

- Artificial Intelligence
- Economics
- Data Science
- Cyber Security
- Business Strategy
- Network Routing

Application of Game Theory in Artificial Intelligence

Artificial Intelligence enables machines to make intelligent decisions without human intervention.

Scenario

Two autonomous vehicles arrive at an intersection simultaneously.

- Each vehicle can either **Go** or **Wait**.
- If both move simultaneously, a collision may occur.
- If both wait, unnecessary delay is introduced.
- The best decision depends on the action of the other vehicle.

Key Observation

Traditional AI answers:

“What is the best action for me?”

Game Theory asks:

“What is the best action considering what the other player may do?”

Game Theory in Artificial Intelligence

Game Theory models autonomous vehicles as rational players.

Car A	Car B	Outcome
Go	Go	Collision
Go	Wait	Safe
Wait	Go	Safe
Wait	Wait	Delay

Each autonomous vehicle predicts the possible actions of the other vehicle and selects the strategy that maximizes safety and efficiency.

Applications

- Multi-Agent Systems
- Autonomous Vehicles
- Robot Collaboration
- Drone Coordination
- AI Negotiation
- Reinforcement Learning

Game Theory in Economics

Game Theory analyzes how individuals, firms, and governments make strategic decisions in competitive environments.

Scenario

- Two competing companies:
 - Company A (Coca-Cola)
 - Company B (Pepsi)
- Both companies have two strategies:
 - Reduce Price
 - Keep Price
- The profit of each company depends on the competitor's pricing decision.

Company A	Company B	Outcome
Reduce Price	Reduce Price	Lower Profit
Reduce Price	Keep Price	A Gains Market Share
Keep Price	Reduce Price	B Gains Market Share
Keep Price	Keep Price	Stable Pricing

Game Theory in Economics (Contd.)

Strategic Decision Making

- Each company predicts the competitor's pricing strategy.
- The selected strategy aims to maximize profit.
- The final outcome depends on the decisions of both companies.

Applications

- Pricing Strategies
- Market Competition
- Auctions
- Stock Markets
- Trade Agreements
- Tax Policy Analysis

Game Theory in Data Science

Data Science uses Game Theory to model strategic interactions among multiple decision-makers and optimize data-driven decisions.

Scenario

- Multiple companies compete for the same online advertisement slot.
- Each company submits a bid based on its advertising budget.
- The advertising platform allocates the slot to the winning bidder.
- Every bidder aims to maximize visibility while minimizing cost.

Company	Bid Amount
Company A	20
Company B	25
Company C	40
Company D	35

Game Theory in Data Science (Contd.)

Strategic Decision Making

- Each advertiser predicts competitors' bidding strategies.
- The platform allocates resources to maximize revenue and relevance.
- Game Theory identifies optimal bidding and allocation strategies.

Applications

- Online Advertising
- Recommendation Systems
- Resource Allocation
- Online Auctions
- Federated Learning
- Multi-Agent Machine Learning

Game Theory in Cyber Security

Cyber Security uses Game Theory to analyze the strategic interaction between attackers and defenders in protecting computer systems and networks.

Scenario

- Two players participate in the security game:
 - Attacker
 - Defender
- The attacker attempts to compromise the system.
- The defender aims to prevent or minimize attacks.
- The success of each player depends on the strategy adopted by the other.

Attacker	Defender	Outcome
Phishing	Firewall	Partial Protection
Malware	Antivirus	Malware Blocked
Ransomware	Encryption	Data Protected
Password Attack	Strong Authentication	Access Denied

Game Theory in Cyber Security (Contd.)

Role of Game Theory

- Predicts the possible actions of attackers and defenders.
- Identifies optimal defense strategies.
- Assists in efficient allocation of security resources.
- Reduces security risks by anticipating future attacks.

Applications

- Intrusion Detection Systems
- Malware Analysis
- Cyber Defense
- Network Protection
- Risk Assessment
- Security Resource Allocation

Game Theory in Business Strategy

Game Theory helps organizations make strategic business decisions by considering the actions and responses of their competitors.

Scenario

- Two competing companies: Amazon and Flipkart
- Both companies decide whether to:
 - Launch a Discount Sale
 - Maintain Regular Prices
- The profit of each company depends on the competitor's strategy.

Amazon	Flipkart	Outcome
Discount Sale	Discount Sale	Reduced Profit for Both
Discount Sale	Regular Price	Amazon Gains Customers
Regular Price	Discount Sale	Flipkart Gains Customers
Regular Price	Regular Price	Stable Competition

Game Theory in Business Strategy (Contd.)

Role of Game Theory

- Predicts competitors' business decisions.
- Identifies optimal pricing and marketing strategies.
- Supports strategic planning and decision making.
- Improves long-term competitive advantage.

Applications

- Pricing Strategies
- Product Launch Planning
- Marketing Campaigns
- Supply Chain Management
- Business Negotiations
- Market Competition Analysis

Game Theory in Network Routing

Game Theory optimizes data transmission by enabling network devices to make strategic routing decisions.

Scenario

- Multiple routers transmit data from a source to a destination.
- Each router selects the shortest or least congested path.
- If all routers choose the same path, network congestion increases.
- Efficient routing requires considering the decisions of other routers.

Router	Selected Path	Result
Router A	Path 1	Congested
Router B	Path 1	Congested
Router C	Path 2	Less Congested
Router D	Path 3	Optimal Routing

Game Theory in Network Routing (Contd.)

Role of Game Theory

- Predicts the routing decisions of other network nodes.
- Distributes network traffic efficiently.
- Reduces congestion and communication delay.
- Improves network performance and resource utilization.

Applications

- Internet Routing
- Wireless Networks
- 5G Communication
- Cloud Computing
- Data Centers
- Sensor Networks

What is a Game?

A **game** is a strategic situation involving two or more decision-makers (players), where each player selects an action (strategy), and the final outcome depends on the combination of all players' decisions.

Characteristics of a Game

- Two or more players participate.
- Every player has one or more possible strategies.
- Each player aims to maximize their own objective.
- The outcome depends on the actions of all players.
- Every outcome is associated with a payoff.

Basic Elements of a Game

Every game consists of four fundamental elements that define the strategic interaction among rational players.



Figure 1: Basic Elements of a Game

Every strategic game is defined by these four fundamental elements.

Relationship among the Components of a Game

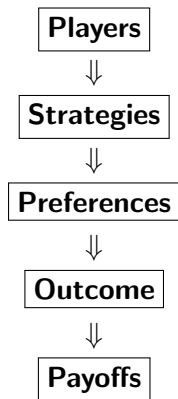


Figure 2: Relationship among the Components of a Game

Game Flow

Players choose strategies based on their preferences.

The interaction of all players' strategies determines the outcome.

Each player receives a payoff corresponding to that outcome.

Players

A **player** is a decision-maker who participates in a game by choosing strategies to achieve a specific objective.

Key Characteristics

- Participates in strategic decision-making.
- May be an individual, organization, or AI agent.
- Selects strategies based on available information.
- Considers other players' possible actions.
- Attempts to maximize own payoff.
- Every game involves two or more players.

Examples

Typical Players

- Robot A vs Robot B
- Amazon vs Flipkart
- Attacker vs Defender
- Car A vs Car B
- Player A vs Player B

Strategies

A **strategy** is a complete plan of action that a player chooses to achieve the best possible outcome in a game.

Key Characteristics

- Represents the set of actions available to a player.
- A player selects one strategy from the available alternatives.
- The chosen strategy influences the final outcome.
- Selection depends on the player's objective and opponents' actions.
- Strategies may be **cooperative** or **competitive**.

Examples

Typical Strategies

- Go / Wait
- Attack / Defend
- Increase / Reduce Price
- Bid High / Bid Low
- Cooperate / Compete

Preferences

Preferences represent the order in which a player ranks the possible outcomes of a game based on their desirability.

Key Characteristics

- Reflect the player's objectives and priorities.
- Indicate which outcomes are more or less preferred.
- Help evaluate the consequences of different strategies.
- Vary from one player to another based on individual goals.
- Form the basis for assigning payoffs in a game.

Examples

Preference Orders

- Safe Arrival $>$ Delay $>$ Accident
- High Profit $>$ Moderate Profit $>$ Loss
- Win $>$ Draw $>$ Lose
- High Accuracy $>$ Low Accuracy
- Low Cost $>$ High Cost

Payoffs

A **payoff** is the numerical value that represents the reward, benefit, or loss received by a player after all players have chosen their strategies.

Key Characteristics

- Quantifies the outcome of a player's strategic decision.
- Represents the player's level of satisfaction or utility.
- Can be positive (gain), negative (loss), or zero (neutral).
- Depends on the combination of strategies chosen by all players.
- Used to compare outcomes and identify the optimal strategy.

Examples

Typical Payoffs

- +10 (Safe & Fast)
- +5 (Safe with Delay)
- 0 (No Gain/No Loss)
- -100 (Collision)
- 50 Cr Profit / 10 Cr Loss

Difference between Ordinary Decision Making and Strategic Decision Making

Ordinary Decision Making	Strategic Decision Making
Decision depends only on your own actions.	Decision depends on your actions and those of others.
Single decision-maker.	Two or more decision-makers (players).
No strategic interaction.	Strategic interaction exists.
Outcome depends only on your decision.	Outcome depends on all players' decisions.
Decisions are made independently.	Decisions anticipate others' actions.
No need to predict others' strategies.	Requires predicting others' strategies.

Case Study: Amazon vs Flipkart

Scenario

During the Diwali sale, **Amazon** and **Flipkart** independently decide whether to:

- Offer **Heavy Discounts**
- Maintain **Regular Prices**

Activity for Students

Identify the following game components:

- Players
- Strategies
- Preferences
- Payoffs

Possible Business Outcomes

- Increased sales
- Higher profit
- Larger market share
- Customer acquisition

Question: Which pricing strategy would maximize the overall payoff for each company?

Why Mathematics in Game Theory?

Strategic decision-making becomes challenging when the outcome of one player's decision depends on the decisions of other players.

Mathematics helps to:

- Represent complex situations systematically.
- Analyze strategic choices objectively.
- Predict possible outcomes.
- Support logical and rational decision-making.

Example:

Pricing decisions of Amazon and Flipkart.

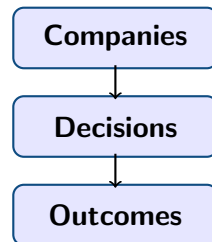


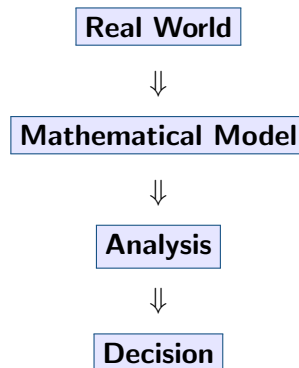
Figure 3: Mathematical Modeling of Strategic Decision-Making

Mathematical Modelling

A **mathematical model** is a simplified representation of a real-world problem using mathematical symbols, variables, equations, and relationships.

Purpose of Mathematical Modelling

- Simplifies complex real-world situations.
- Focuses on important factors affecting decisions.
- Helps analyze strategic interactions.
- Predicts possible outcomes.



Why Do We Use Mathematical Models?

Real-world problems contain a large amount of information. A mathematical model considers only the information that is necessary for effective decision-making.

Example: Amazon vs. Flipkart

Information Ignored

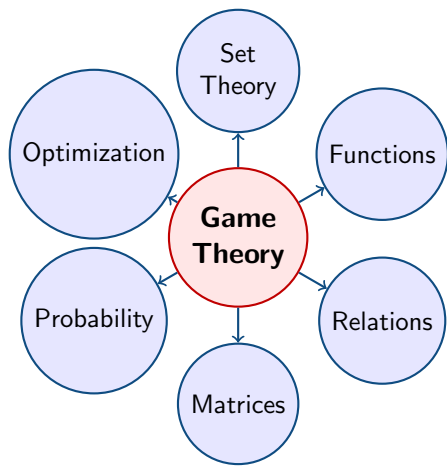
- Office location
- CEO
- Number of employees
- Company logo
- Employee salary

Information Considered

- Players
- Strategies
- Preferences
- Payoffs

Key Idea: A mathematical model ignores unnecessary details and focuses only on information relevant for strategic decision-making.

Mathematical Concepts Used in Game Theory



Game Theory employs several mathematical concepts to model and analyze strategic interactions among rational players.

Purpose

- Represent strategic situations mathematically.
- Model interactions among players.
- Analyze possible outcomes.
- Predict rational decisions.
- Support optimal strategy selection.

Figure 5: Mathematical Concepts Used in Game Theory

Set Theory in Game Theory

A **set** is a well-defined collection of distinct objects. In Game Theory, sets are used to represent the fundamental components of a game.

Sets are used to represent

- Players
- Strategies
- Outcomes

Example

Set of Players

$$N = \{\text{Amazon, Flipkart}\}$$

Set of Strategies

$$S = \{\text{Reduce Price, Keep Price}\}$$

Observation: Every game is mathematically described by defining sets of players, available strategies, and possible outcomes.

Cartesian Product

The **Cartesian Product** represents all possible combinations of strategies chosen by the players.

Example

Amazon's Strategy Set

$$S_A = \{\text{Reduce Price, Keep Price}\}$$

Flipkart's Strategy Set

$$S_B = \{\text{Reduce Price, Keep Price}\}$$

Cartesian Product

$$S_A \times S_B$$

Possible Strategy Combinations

(Reduce, Reduce)

(Reduce, Keep)

(Keep, Reduce)

(Keep, Keep)

Functions and Payoff Functions

A **function** maps an input to an output. In Game Theory, a payoff function maps the strategy profile chosen by the players to the corresponding payoffs.

Input

- Strategies selected by the players
- One strategy from each player's strategy set

Output

- Payoffs received by each player
- Numerical reward or utility

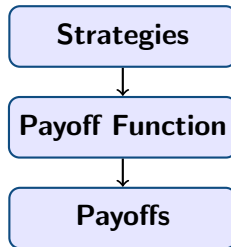
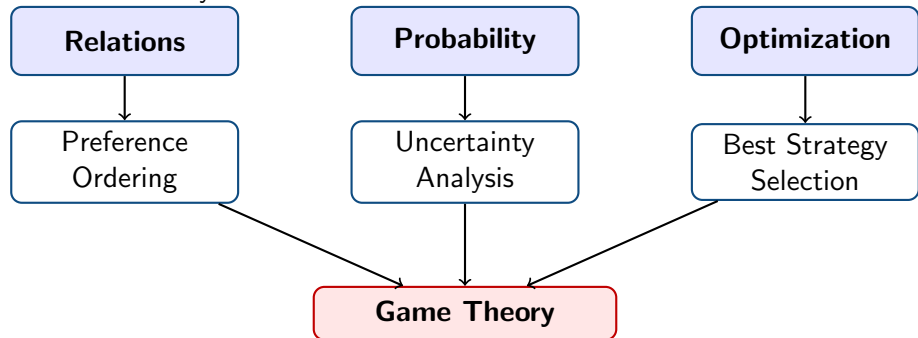


Figure 6: Mapping from Strategies to Payoffs

Relations, Probability and Optimization

These mathematical concepts provide the mathematical foundation for analyzing strategic interactions in Game Theory.



- **Relations** model players' preferences over possible outcomes.
- **Probability** captures uncertainty and mixed strategies.
- **Optimization** identifies strategies that maximize payoff or minimize loss.

Relations in Game Theory

Relations represent a player's preference ordering over different possible outcomes.

Role of Relations

- Compare different outcomes.
- Rank outcomes from most preferred to least preferred.
- Model players' preferences mathematically.
- Help analyze rational decision making.

Example

High Profit
>
Moderate Profit
>
Loss

Think Before We Learn: Optimization

Before defining **Optimization**, let's answer the following questions.

- ① You have three routes to college taking **25, 35, and 45 minutes**. Which route would you choose? Why?
- ② You trained three Machine Learning models with accuracies of **89%, 92%, and 95%**. Which model would you deploy? Why?
- ③ A delivery robot can reach its destination using three paths requiring **8, 12, and 18 minutes**. Which path should it select? Why?
- ④ You receive two job offers with different salaries and career growth opportunities. Which offer would you accept? Why?

What is the common objective behind all these decisions?

Think Before We Learn: Rational Choice

Before defining **Rational Choice**, answer the following questions.

- ① You have Rs. 50,000 to buy a laptop. Which laptop would you choose? Why?
- ② You trained three Machine Learning models with accuracies of 89%, 92%, and 95%. Which model would you deploy? Why?
- ③ Google Maps suggests three routes taking 20, 30, and 45 minutes. Which route would you choose? Why?
- ④ Two job offers provide different salaries and career growth opportunities. Which offer would you accept? Why?

How did you decide which option to choose?

Rational Choice

A **Rational Choice** is the selection of the best available alternative that helps achieve a specific objective based on the available information.

A rational decision-maker

- Identifies all available alternatives.
- Evaluates the advantages and disadvantages of each alternative.
- Considers the available information.
- Selects the alternative that provides the maximum benefit or minimum cost.

Examples

- Choosing the fastest route to college.
- Selecting the Machine Learning model with the highest accuracy.
- Buying a laptop with the best specifications within the budget.

Rational Choice in Game Theory

In Game Theory, a player's decision depends not only on their own objective but also on the expected actions of other players.

Example

Two autonomous vehicles reach an intersection.

- Each vehicle can either **Go** or **Wait**.
- Choosing **Go** immediately may cause a collision.
- Choosing **Wait** may introduce a delay.
- Each vehicle predicts the other's action before making a decision.

The selected strategy that maximizes safety and efficiency is the **rational choice**.

Characteristics of Rational Choice

A rational choice has the following characteristics:

- Clearly defined objective.
- Multiple alternatives to choose from.
- Evaluation of available information.
- Comparison of possible outcomes.
- Selection of the best available alternative.
- Maximization of payoff or minimization of loss.

Important Note

A rational choice does not always guarantee success. It represents the best possible decision based on the available information at that time.

Why Do We Need Solution Concepts?

Consider two competing companies deciding whether to offer discounts during a festive sale.

Possible Strategies

- Company A: Discount or Regular Price
- Company B: Discount or Regular Price

Strategy Combination	Possible Outcome
(Discount, Discount)	Lower Profit for Both
(Discount, Regular)	Company A Gains Customers
(Regular, Discount)	Company B Gains Customers
(Regular, Regular)	Stable Competition

Which strategy combination should the companies choose?

Basic Solution Concepts

A **solution concept** is a rule or principle used to determine the expected outcome of a game when all players make rational decisions.

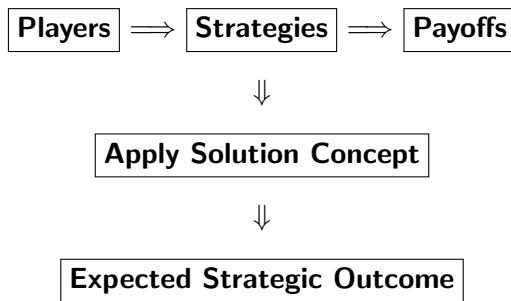
Why is it required?

- Multiple players have different objectives.
- Several strategy combinations are possible.
- Every player tries to maximize their own payoff.
- A solution concept helps identify the most appropriate strategic outcome.

Players $\rightarrow$ Strategies $\rightarrow$ Payoffs $\rightarrow$ Solution Concept $\rightarrow$ Expected Outcome

How is a Solution Obtained?

A solution concept determines the expected outcome of a game by systematically analyzing the decisions made by all players.



The solution concept evaluates all possible strategy combinations and predicts the most appropriate outcome when players act rationally.

Types of Solution Concepts

Different games require different solution concepts to determine the expected outcome.

Solution Concept	Description
Dominant Strategy	Best strategy regardless of the opponent's action.
Nash Equilibrium	No player benefits by changing strategy alone.
Pareto Optimality	Improves one player's payoff only by reducing another's payoff.
Maximin Strategy	Selects the strategy with the best worst-case outcome.

Why Do We Need a Dominant Strategy?

In many games, a player has multiple strategies available.

The challenge is to determine:

- Which strategy should the player choose?
- Does the best strategy depend on the opponent's decision?
- Is there a strategy that is always better than the others?

Game Theory answers these questions using the concept of a **Dominant Strategy**.

Understanding Dominance

A strategy is said to **dominate** another strategy if it performs better in every possible situation.

Situation	Better Strategy
Situation 1	Strategy A
Situation 2	Strategy A
Situation 3	Strategy A
Situation 4	Strategy A

Observation

- Strategy A is preferred in every situation.
- Strategy B is never the better choice.
- Strategy A dominates Strategy B.
- Strategy A is the dominant strategy.

Dominant Strategy

Definition

- A **Dominant Strategy** is a strategy that gives a player the highest payoff regardless of the strategy chosen by the other player.

Characteristics

- Always provides the highest payoff.
- Independent of the opponent's decision.
- Simplifies strategic decision-making.
- May exist for one player, both players, or neither player.

Key Idea

- If one strategy is always better than every alternative, a rational player will choose that strategy.

How to Identify a Dominant Strategy?

To determine whether a dominant strategy exists, follow these steps.

- 1 Select one player at a time.
- 2 Compare the player's payoffs for each possible strategy of the opponent.
- 3 Identify whether one strategy always provides a better payoff.
- 4 Repeat the same process for the other player.
- 5 If one strategy is always better, it is the **Dominant Strategy**.

Note

- Compare only one player's payoffs at a time.
- Do not compare Player A's payoff with Player B's payoff.

Example: Identifying a Dominant Strategy

Consider the following payoff matrix. The first value in each cell represents the payoff of **Player A**, and the second value represents the payoff of **Player B**.

	Left	Right
Up	(8,6)	(6,4)
Down	(5,5)	(3,2)

Task

- Compare Player A's payoffs.
- Compare Player B's payoffs.
- Identify the dominant strategy for each player.

Solution: Dominant Strategy

Player A

- If Player B chooses Left: $8 > 5$
- If Player B chooses Right: $6 > 3$
- Therefore, **Up** is the dominant strategy.

Player B

- If Player A chooses Up: $6 > 4$
- If Player A chooses Down: $5 > 2$
- Therefore, **Left** is the dominant strategy.

Conclusion

- Dominant Strategy for Player A: **Up**
- Dominant Strategy for Player B: **Left**

Result of the Game

From the previous analysis,

- Player A's dominant strategy is **Up**.
- Player B's dominant strategy is **Left**.
- Both players will independently choose their dominant strategies.

Player	Dominant Strategy
Player A	Up
Player B	Left

Final Strategy Profile:

(Up, Left)

Does Every Game Have a Dominant Strategy?

A dominant strategy does not exist in every game.

Possible Situations

- Both players have dominant strategies.
- Only one player has a dominant strategy.
- Neither player has a dominant strategy.

Observation

- The existence of a dominant strategy depends on the payoff values.
- If no dominant strategy exists, other solution concepts are required.

The most widely used solution concept is Nash Equilibrium.

Dominated Strategy

Definition

- A **Dominated Strategy** is a strategy that always provides a lower payoff than another available strategy.
- A rational player will never choose a dominated strategy.

Characteristics

- Always performs worse than another strategy.
- Can be eliminated from the analysis.
- Simplifies the process of solving a game.
- May be dominated by one or more strategies.

Example of a Dominated Strategy

Consider the following payoff matrix.

Player A	Player B	
	Left	Right
Up	(8,5)	(6,4)
Down	(5,3)	(3,2)

Observation

- Compare Player A's payoffs.
- Down gives a lower payoff in every case.
- Therefore, **Down** is the dominated strategy.

Dominant vs Dominated Strategy

Dominant Strategy	Dominated Strategy
Always gives a better payoff.	Always gives a lower payoff.
Preferred by a rational player.	Avoided by a rational player.
Selected during decision-making.	Eliminated during analysis.
May or may not exist.	May or may not exist.

Need for Nash Equilibrium

Observation

- Dominant Strategy provides a simple solution when it exists.
- However, not every game has a dominant strategy.
- In many games, the best strategy depends on the opponent's decision.
- Such games cannot be solved using the Dominant Strategy concept alone.

Conclusion

- A more general solution concept is required.
- This leads to the concept of **Nash Equilibrium**.

Why Do We Need Nash Equilibrium?

Observation

- Some games do not have a dominant strategy.
- The best strategy may change according to the opponent's decision.
- Players cannot identify the best strategy independently.
- The final decision depends on the strategies chosen by all players.

Question

- How can rational players determine the final outcome?
- Which strategy combination should the players choose?

Game Theory answers this using the concept of [Nash Equilibrium](#).

Nash Equilibrium

Definition

- A **Nash Equilibrium** is a strategy profile in which no player can improve their payoff by changing only their own strategy while the other players keep their strategies unchanged.

Key Characteristics

- Every player chooses the best response.
- No player has an incentive to change their strategy alone.
- The outcome remains stable.
- One or more Nash Equilibria may exist.

Best Response

Definition

- A **Best Response** is the strategy that gives a player the highest payoff for a given strategy chosen by the opponent.

Key Points

- Depends on the opponent's strategy.
- Maximizes the player's payoff.
- Different opponent strategies may lead to different best responses.
- Forms the foundation of Nash Equilibrium.

Observation

- Unlike a Dominant Strategy, a Best Response may change when the opponent changes their strategy.

Identifying the Best Response

Consider the following payoff matrix.

Player A	Player B	
	Left	Right
Up	(4,3)	(2,5)
Down	(5,2)	(3,4)

Observation

- If Player B chooses **Left**, Player A's best response is **Down**.
- If Player B chooses **Right**, Player A's best response is **Down**.
- The best response is identified by comparing Player A's payoffs.

Finding the Nash Equilibrium

A **Nash Equilibrium** occurs when both players simultaneously choose their best responses.

Player A	Player B	
	Left	Right
Up	(4,3)	(2,1)
Down	(3,2)	(5,4)

Legend

- Blue : Best Response of Player A
- Orange : Best Response of Player B
- Green Cell : Nash Equilibrium

Why is Nash Equilibrium Stable?

At Nash Equilibrium

- Every player is playing their best response.
- No player can increase their payoff by changing only their own strategy.
- Any unilateral change leads to an equal or lower payoff.
- Therefore, no player has an incentive to deviate.

Result

- The strategy profile remains unchanged.
- The game reaches a stable outcome called the **Nash Equilibrium**.

Example of Nash Equilibrium

Consider the following payoff matrix.

Player A	Player B	
	Left	Right
Up	(3,2)	(1,4)
Down	(2,1)	(4,3)

Observation

- Both highlighted cells satisfy the Nash Equilibrium condition.
- Neither player can improve their payoff by changing only their own strategy.

Applications of Nash Equilibrium

Nash Equilibrium is widely used to analyze strategic decision-making in real-world scenarios.

Applications

- Pricing strategies in business competition.
- Autonomous vehicle decision-making.
- Online auctions and bidding systems.
- Cyber security and defense strategies.
- Resource allocation in communication networks.
- Multi-agent systems and Artificial Intelligence.

Observation

- Nash Equilibrium helps predict stable outcomes when multiple decision-makers interact strategically.

Why Do We Need Pareto Optimality?

In many games, multiple outcomes are possible.

Question

- Is the selected outcome beneficial for all players?
- Can one player's payoff be improved without reducing another player's payoff?
- How can we identify the most efficient outcome?

Game Theory answers these questions using the concept of **Pareto Optimality**.

Pareto Optimality

Definition

- An outcome is **Pareto Optimal** if no player's payoff can be increased without decreasing another player's payoff.

Characteristics

- Represents an efficient outcome.
- Improves one player's payoff only by reducing another player's payoff.
- No further mutual improvement is possible.

Understanding Pareto Optimality

Consider the following outcomes.

Outcome	Player A	Player B
A	2	2
B	4	2
C	4	5
D	5	3

Observation

- Outcome A can be improved.
- Outcome C provides higher payoffs to both players.
- Therefore, Outcome A is not Pareto Optimal.

Identifying Pareto Optimality

Consider Outcome C.

$$\text{Player A} = 4 \qquad \text{Player B} = 5$$

Observation

- Player A's payoff cannot be increased without reducing Player B's payoff.
- Player B's payoff cannot be increased without reducing Player A's payoff.
- Therefore, Outcome C is **Pareto Optimal**.

Nash Equilibrium vs Pareto Optimality

Nash Equilibrium	Pareto Optimality
Stable outcome	Efficient outcome
No player changes strategy alone	No player can be made better off without making another worse off
Focuses on strategic stability	Focuses on overall efficiency

Why Do We Need Maximin Strategy?

In some games, players may not know the opponent's strategy.

Question

- What if the opponent chooses the worst possible action?
- Which strategy protects a player from the maximum possible loss?
- How can a player make a safe decision under uncertainty?

Game Theory addresses this using the concept of **Maximin Strategy**.

Maximin Strategy

Definition

- A **Maximin Strategy** selects the strategy whose worst possible payoff is the highest.

Characteristics

- Focuses on the worst-case outcome.
- Reduces the risk of large losses.
- Suitable when the opponent's strategy is unknown.
- Commonly used in competitive and adversarial situations.

Understanding Maximin Strategy

Consider the following payoff matrix for Player A.

Strategy	Left	Right
Up	3	6
Down	4	5

Observation

- Worst payoff for Up = 3
- Worst payoff for Down = 4
- Choose the strategy with the higher worst payoff.

Identifying the Maximin Strategy

From the previous payoff matrix,

Strategy	Minimum Payoff
Up	3
Down	4

Conclusion

- The minimum payoff of Up is 3.
- The minimum payoff of Down is 4.
- Since 4 is greater than 3, **Down** is the Maximin Strategy.

Applications of Maximin Strategy

Maximin Strategy is commonly used in:

- Cyber security for defensive planning.
- Military and defense decision-making.
- Financial risk management.
- Artificial Intelligence and adversarial games.
- Resource allocation under uncertainty.

Key Idea

- Maximin Strategy helps a player secure the best possible outcome even in the worst-case scenario.

Why Do We Need Different Types of Games?

Not all games are played in the same way.

Question

- Can players communicate and make agreements?
- Do players always act independently?
- How does cooperation influence the outcome of a game?

Game Theory classifies games into **Cooperative** and **Non-Cooperative** Games.

Cooperative Games

Definition

- A **Cooperative Game** is a game in which players can communicate, negotiate, and form binding agreements to achieve a common objective.

Characteristics

- Players cooperate with each other.
- Agreements are binding.
- Resources and rewards may be shared.
- Focuses on maximizing the group's overall benefit.

Non-Cooperative Games

Definition

- A **Non-Cooperative Game** is a game in which players make decisions independently without forming binding agreements.

Characteristics

- Players act independently.
- No binding agreements exist.
- Each player maximizes their own payoff.
- Strategic decisions depend on the actions of other players.

Cooperative vs Non-Cooperative Games

Cooperative Games	Non-Cooperative Games
Players can cooperate.	Players act independently.
Binding agreements exist.	No binding agreements.
Focus on group benefit.	Focus on individual benefit.
Rewards may be shared.	Each player keeps their own payoff.

Examples of Cooperative and Non-Cooperative Games

Cooperative Games

- Business partnerships
- Team projects

Non-Cooperative Games

- Prisoner's Dilemma
- Chess
- Online auctions
- Price competition between companies

Mathematical Representation of Games

Cooperative Game

$$G = (N, v)$$

- N : Set of players.
- v : Coalition value (worth) function.
- Players cooperate to maximize the coalition's payoff.

Non-Cooperative Game

$$G = (N, S, U)$$

- N : Set of players.
- S : Strategy set of each player.
- U : Utility (payoff) function.

Coalition Formation

Coalition

- A coalition is a group of players who cooperate to achieve a common objective.
- Players coordinate their strategies for mutual benefit.
- The coalition receives a collective payoff.

Benefits

- Resource sharing
- Improved decision-making
- Increased collective payoff
- Reduced individual risk

Communication Between Agents

Cooperative Games

- Players exchange information.
- Strategies are coordinated.
- Decisions are made collaboratively.

Non-Cooperative Games

- Players act independently.
- No information sharing.
- Decisions are based on individual objectives.